

Bhaskara II, Bijaganita, parts 1-3

English Translation

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

About this reader

- Reader type - English Translation.
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BILINGUAL EDITION

Sanskrit – English / Saṃskṛta – Aṅgla

बीजगणितमध्यगणितं वा

BIJAGANITA

(Seed Computation / Elements of Algebra)

श्रीभास्कराचार्यविरचितम् ।

Composed by Śrī Bhāskarācārya

Part 1 / Prathamo Bhāgaḥ

Edited by Sudhākara Dvivedī with further notes by Muralīdhara Jha

Benares Sanskrit Series, No. 159 (1927)

Bilingual edition with IAST transliteration following S. K. Abhyankar's conventions

Translator's Preface

This bilingual edition presents Bhāskara II's *Bījagaṇita* ("Seed Computation") in parallel Sanskrit-English columns. The text is based on the Benares Sanskrit Series edition of 1927, edited by Sudhākara Dvivedī with additional notes by Muralīdhara Jha.

Key Terminology:

- बीज (*bīja*) — seed, origin; hence *algebra*
- अव्यक्त (*avyakta*) — unseen, unknown quantity
- यावत्तावत् (*yāvat-tāvat*) — "as much as," the unknown x
- रूप (*rūpa*) — absolute number, constant term
- वर्ण (*varṇa*) — colour, representing different unknowns
- करणी (*karaṇī*) — surd, irrational quantity
- समीकरण (*samīkaraṇa*) — equation (making equal)
- धन (*dhana*) — positive (wealth)
- ऋण (*ṛṇa*) — negative (debt)

The left column presents the original Sanskrit in Devanāgarī script. The right column provides the English translation with IAST transliteration of key terms.

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श्रीगणेशाय नमः

Obeisance to Śrī Gaṇeśa

अथ बीजगणितम्

Now begins the Bījagaṇita

उपोद्घातः

Upodghāta (Introduction)

श्रीं ॐ मते बलवते हनुमते नमः ।

उत्पादकं यत् प्रवदन्ति बुद्धेः स्थितं सुरैरपि ।
अस्य क्रिया तत्कथयाम्यथवा तद्वीजमष्टधा ॥

पूर्वं प्रोक्तं व्यक्तमस्य कथये प्रज्ञा ते च मेऽव्यक्तयुक्तया ।
यं तु शक्या मन्दबुद्धिर्न तत्समं न तद् द्वयं वीजक्रिया च ॥

Obeisance to the powerful Hanūmat, the abode of wisdom.

That which the sages declare as the source of intellect,
even by the gods — its eight-fold *bīja* operation I shall tell.

First the manifest (*vyakta*) was spoken; now I speak the
unmanifest (*avyakta*)
to those of wisdom; the slow-witted cannot grasp this
two-fold *bīja*-operation.

:

Explanation: Bhāskara states that he will explain the
eight-fold operations of algebra (*bījagaṇita*). Those
accomplished in manifest arithmetic (*vyakta-gaṇita*) should
also study the unmanifest (*avyakta*). Algebra is the root of all
computation.

:

Fundamental Concepts:

- —
- (x) —
- , , , —
- —
- () (-) —
- —

- *Avyakta-kalpanā* — representation of unknown quantities
- *Yāvat-tāvat* (x) — first unknown quantity
- *Kālaka*, *Nīlaka*, *Pītaka*, *Haritaka* — other colours for unknowns
- *Rūpa* — absolute constants
- *Dhana* (+) and *ṛṇa* (-) — signs
- *Karaṇī* — irrational (surd) quantities

ब्रह्मस्फुटसिद्धान्ते च कालक्रियायां तथैव च ।
ग्रहगणितं निगदितं त्रैशिकफलं तथा ॥

In the *Brāhmasphuṭasiddhānta* and in *Kālakriyā*,
and *Graha-gaṇita* and the Rule of Three — all was told.

घनाणां सङ्कलनम्

Ghanāṇām Saṅkalanam (Sum of Cubes)

घनाणां सङ्कलने करणसूत्रं बुद्धये

Sūtra for Sum of Cubes

घनाः समं सङ्गुणिता द्वित्रिभिः समन्विताः
एकघनाद्विवर्जितास्ते सङ्कलिताः स्युः सदा घनाः ॥ १ ॥

*ghanāḥ samaṁ saṅguṇitā dvitribhiḥ samanvitāḥ
ekaghanād vivarjitās te saṅkalitāḥ syuḥ sadā ghanāḥ —*
Verse 1

:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

:

$$1^3 + 2^3 + 3^3 + 4^3 + 5^3 = \left(\frac{5 \cdot 6}{2} \right)^2$$

$$= 15^2 = 225$$

$$: 1 + 8 + 27 + 64 + 125 = 225$$

Rule: The sum of cubes of the first n natural numbers equals the square of their sum:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Example: Sum of the first 5 cubes:

$$1^3 + 2^3 + 3^3 + 4^3 + 5^3 = \left(\frac{5 \cdot 6}{2} \right)^2$$

$$= 15^2 = 225$$

Verification: $1 + 8 + 27 + 64 + 125 = 225 \checkmark$

घनानां व्यवकलनम्

Ghanānām Vyavakalanam (Subtraction of Cubes)

व्यस्तद्वयोर्घनयोर्व्यस्तयोर्घनयोर्व्यवकलनम् ।
समघनयोरपि तथा व्यवकलनं सम्भवत्येव ॥ २ ॥

*vyastadvayor ghanayor vyastayor ghanayor vyavakalanam
samaghnayor api tathā vyavakalaṁ sambhavaty eva — Verse
2*

: $a - b$,

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Rule: The subtraction of cubes of two quantities equals the product of their difference and the sum of their squares plus their product:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

:

$$(yā 5)^3 - (yā 2)^3 = 125yā^3 - 8yā^3 \\ = 117yā^3$$

Example:

$$(5yā)^3 - (2yā)^3 = 125yā^3 - 8yā^3 \\ = 117yā^3$$

$$: (5 - 3)(25 + 15 + 9) = 2 \cdot 49 = 98 ()$$

Or: $(5 - 3)(25 + 15 + 9) = 2 \cdot 49 = 98$ (in constants)

घनानां गुणनम्

Ghanānām Guṇanam (Multiplication of Cubes)

घनयोर्घनयोर्घनयोर्वा गुणनं सम्भवति ।
घनघातः सम्भवत्येवं गुणकारघनयोः समं ॥ ३ ॥

ghanayor ghanayor ghanayor vā guṇanam sambhavati
ghanaghātaḥ sambhavaty evaṁ guṇakāraghanayoḥ samaṁ
— Verse 3

:

$$a^3 \times b^3 = (ab)^3$$

Rule: The product of two cubes equals the cube of the product of their roots:

$$a^3 \times b^3 = (ab)^3$$

:

$$(3y\bar{a})^3 \times (2y\bar{a})^3 = (6y\bar{a})^3 \\ = 216y\bar{a}^3$$

Example:

$$(3y\bar{a})^3 \times (2y\bar{a})^3 = (6y\bar{a})^3 \\ = 216y\bar{a}^3$$

घनानां भागहारः

Ghanānām Bhāgahārah (Division of Cubes)

घनं घनेन विभजेद् घनमूलयोर्मूलयोरिव ।
घनभागः स्तः सम्भवत्येवं भागकारयोः समं ॥ ४ ॥

*ghanam ghanena vibhajed ghanamūlayor mūlayor iva
ghanabhāgaḥ sthaḥ sambhavaty evaṁ bhāgakārayoḥ samam*
— Verse 4

:

$$\frac{a^3}{b^3} = \left(\frac{a}{b}\right)^3$$

:

$$\begin{aligned} \frac{(8y\bar{a})^3}{(2y\bar{a})^3} &= \left(\frac{8y\bar{a}}{2y\bar{a}}\right)^3 \\ &= (4y\bar{a})^3 = 64y\bar{a}^3 \end{aligned}$$

Rule: A cube divided by a cube; the quotient is the cube of the quotient of their roots:

$$\frac{a^3}{b^3} = \left(\frac{a}{b}\right)^3$$

Example:

$$\begin{aligned} \frac{(8y\bar{a})^3}{(2y\bar{a})^3} &= \left(\frac{8y\bar{a}}{2y\bar{a}}\right)^3 \\ &= (4y\bar{a})^3 = 64y\bar{a}^3 \end{aligned}$$

घनानां वर्गो मूलं च

Ghanānām Vargo Mūlaṃ ca (Square and Root of Cubes)

घनस्य वर्गः स्यात् षष्ठी घनस्य मूलं तु मूलघनम् ।
वर्गमूलं घनमूलं च कल्पनीयं ततस्ततः ॥ ५ ॥

*ghanasya vargaḥ syāt ṣaṣṭhī ghanasya mūlaṃ tu mūlaghanam
vargamūlaṃ ghanamūlaṃ ca kalpanīyaṃ tatas tataḥ —*
Verse 5

:

$$(a^3)^2 = a^6 \quad (\quad = \quad)$$

$$\sqrt[3]{a^3} = a \quad (\quad)$$

$$\sqrt{a^3} = a^{3/2} \quad (\quad)$$

Rule: The square of a cube is the sixth power. The root of a cube is the cube-root.

$$(a^3)^2 = a^6 \quad (\text{square of cube} = 6\text{th power})$$

$$\sqrt[3]{a^3} = a \quad (\text{cube root})$$

$$\sqrt{a^3} = a^{3/2} \quad (\text{square root of cube})$$

अव्यक्तसङ्कलनव्यवकलनम्

Avyakta-saṅkalana-vyavakalanam
(Addition/Subtraction of Unknowns)

(१) यावत्तावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः पीतो हरितश्चेत् द्वयं च अव्यक्तानां कल्पना मतसंज्ञा—
राशिसंख्यानां कल्पिता मताचार्येण ॥ ६ ॥

(1) *yāvat-tāvat kālaka nīlaka pītaka haritakādayaḥ*
varṇāḥ — The unknown quantities are represented by
colours: *yāvat-tāvat* (x), *kālaka* (y), *nīlaka* (z), *pītaka* (u),
haritaka (v), etc.

(): ($= z$), ($= u$), ($= v$) (): ($= x$), ($= y$),

$$(yā\ 1 + kā\ 1) + (yā\ 2 + kā\ 8) = yā\ 3 + kā\ 9$$

$$(yā\ 5 + kā\ 3) - (yā\ 2 + kā\ 8) = yā\ 3 + kā\ 5$$

Rule (1): Unknown quantities are represented by colours:
yāvat-tāvat (x), *kālaka* (y), *nīlaka* (z), *pītaka* (u), *haritaka*
(v). **Rule (2):** Like terms are combined; unlike terms are set
down separately.

$$(yā\ 1 + kā\ 1) + (yā\ 2 + kā\ 8) = yā\ 3 + kā\ 9$$

$$(yā\ 5 + kā\ 3) - (yā\ 2 + kā\ 8) = yā\ 3 + kā\ 5$$

: “ ” ‘ ’

Note: From the Amarakosha: “*yāvat-tāvat sākalyena*
manasyavasthā” — the term *yāvat-tāvat* means “as much as,”
representing the state of the mind regarding an unknown quantity.

अव्यक्तगुणनम्

Avyakta-guṇanam (Multiplication of Unknowns)

अव्यक्तवर्गविधम्

Avyakta-varga-vidham

अव्यक्ताकृतयोः समस्य हेतोः समेत्य वर्गद्वयं कल्पयेत् ।
तदा तयोर्वर्गयोर्वर्गः सम्भवत्येवमेकस्य वर्गः ॥ ७ ॥

*avyaktākṛtayoh samasya hetoh sametya vargadvayaṃ
kalpayet
tadā tayor vargayor vargaḥ sambhavaty ekasya vargaḥ —
Verse 7*

: वर्गः , ,
:

Rule: When two unknowns are multiplied, a new species is produced. The square of an unknown is *varga*; higher powers follow. **Example:**

$$(yā 5 + kā 3) \times (yā 3 + kā 2) = yāv 15 + yā 19 + kā 6$$

$$: yā 5 kā 3 \quad yā 3 kā 2 = yāv 15 + yā 19 + kā 6$$

$$(5yā + 3kā) \times (3yā + 2kā) = 15yā^2 + 19yā + 6kā$$

With sign change of multiplicand:

$$(5yā - 3kā) \times (3yā + 2kā) = 15yā^2 + yā - 6kā$$

अव्यक्तभागहारः

Avyakta-bhāgahārah (Division of Unknowns)

भागाहरे करणसूत्रं बुद्धये ।
 भाज्यच्छेदः सृजति प्रच्युतः सन् क्षेपे क्षेपे स्थानक्षेपं क्षेपेण ।
 दैन्यैर्द्विजैः सङ्गुणो यैः क्षेपैर्भागाहरे लब्धयस्तः स्पृशेत्

bhāgāhare karaṇasūtram buddhaye
bhājyacchedaḥ sṛjati pracyutaḥ san kṣepe kṣepe
sthānakṣepaṁ kṣepeṇa

— The division of unknown quantities follows the method for known quantities.

:
 : $yā$ 18 $kā$ 24 $kā$ 6 $yā$ 5

: $yāv$ 10 + $yā$ 27 + $yā$ 18 + $kā$ 24

: $yā$ 5 + $kā$ 6

: $yā$ 2 + $yā$ 3 + $kā$ 4

Rule: Division of unknowns is the same as for knowns. If division is not exact, a fraction results. **Example:** Divide $18yā + 24kā$ by $5yā + 6kā$:

Dividend: $10yā^2 + 27yā + 18yā + 24kā$

Divisor: $5yā + 6kā$

Quotient: $2yā + 3yā + 4kā$

अव्यक्तवर्गादि

Avyakta-vargādi (Square etc. of Unknowns)

वर्गः प्रथमं ततो घनोऽथ वर्गवर्गस्ततोऽष्टघातः ।
नवमो वर्गघनश्चेत्यादि घातानां क्रमेण स्मृतम्

*vargaḥ prathamam tato ghano 'tha vargavargas tato
'ṣṭaghātaḥ
navamo vargaghanaś cetyādi ghātānām krameṇa smṛtam*

:

- वर्ग — $yā^2$ (:)
- घन — $yā^3$ (:)
- वर्गवर्ग — $yā^4$ (:)
- घनवर्ग — $yā^5$ (:)
- घनघन — $yā^6$ (:)
- वर्गघन — $yā^6$
- अष्टघात — $yā^8$ (:)
- घनघनवर्ग — $yā^9$ (:)

Powers of Unknowns (in order):

- Varga — $yā^2$ (square)
- Ghana — $yā^3$ (cube)
- Varga-varga — $yā^4$ (square of square)
- Ghana-varga — $yā^5$ (square of cube)
- Ghana-ghana — $yā^6$ (cube of cube)
- Varga-ghana — $yā^6$ (also)
- Aṣṭaghāta — $yā^8$ (8th power)
- Ghana-ghana-varga — $yā^9$ (9th power)

अनेकवर्णीद्विपद्विभ्रम

Aneka-varṇī-dvi-pada-vibhrama

द्विवर्णकं द्विपदं त्रिवर्णकं त्रिपदं तथा ।
बहुवर्णकं बहुपदं च कल्पयेत् सुबुद्धिभिः

*dvivarṇakam dvipadam trivarṇakam tripadam tathā
bahuvarṇakam bahupadam ca kalpayet subuddhibhiḥ*

: (=), (=), , , ,
—
(c + d) ac + ad + bc + bd (a + b)

Rule: Expressions with two unknowns (*dvivarṇaka*), two terms (*dvipada*), three unknowns (*trivarṇaka*), three terms (*tripada*), and many unknowns (*bahuvarṇaka*) or many terms (*bahupada*) should be formed by the intelligent.

Bhāskara explains operations on polynomials. Multiplication of binomials: $(a + b) \times (c + d) = ac + ad + bc + bd$.

करणीसङ्कलनव्यवकलनम्

Karaṇī-saṃkalana-vyavakalanam
(Addition/Subtraction of Surds)

करणीसंयुक्तिरेभिः सङ्कलिता स्यात् समं करणीभिः ।
करणीभिर्विशिष्टाः पृथक् स्थापनीयाः समं करणीभिः

*karaṇīsaṃyuktirebhiḥ saṃkalitā syāt samam karaṇībhiḥ
karaṇībhir viśiṣṭāḥ prthak sthāpanīyāḥ samam karaṇībhiḥ*

$$\begin{aligned} & : \\ \sqrt{3} + \sqrt{3} + 2\sqrt{3} + 5\sqrt{3} & : \\ & : (1 + 1 + 2 + 5)\sqrt{3} = 9\sqrt{3} \\ & : \sqrt{2} + \sqrt{3} + \sqrt{5} \\ & : \sqrt{2} + \sqrt{3} + \sqrt{5} \quad (\quad) \end{aligned}$$

Rule: Surds having the same radical are added together; those with different radicals are set down separately.

Example 1: Add $\sqrt{3} + \sqrt{3} + 2\sqrt{3} + 5\sqrt{3}$.

$$\text{Sum: } (1 + 1 + 2 + 5)\sqrt{3} = 9\sqrt{3}$$

Example 2: Add $\sqrt{2} + \sqrt{3} + \sqrt{5}$.

Result: $\sqrt{2} + \sqrt{3} + \sqrt{5}$ (set down separately)

करणीगुणनम्

Karaṇī-guṇanam (Multiplication of Surds)

करणीगुणने करणीगुणने समे समे विपरे विपरे ।
गुण्यकरणीगुणककरणी गुणनफलं करणी भवति

*karaṇīguṇane karaṇīguṇane same same vipare vipare
guṇyakaraṇīguṇakakaraṇī guṇanaphalaṃ karaṇī bhavati*

$$: 3\sqrt{2} \quad - \quad 4\sqrt{3}$$

$$: (3 \times 4) \times \sqrt{2 \times 3} = 12\sqrt{6}$$

$$: a\sqrt{b} \times c\sqrt{d} = (ac)\sqrt{bd}$$

Rule: In multiplication of surds, multiply the radical quantities to obtain the new radical; multiply the coefficients for the new coefficient. **Example:** Multiply $3\sqrt{2}$ by $4\sqrt{3}$.

$$\text{Product: } (3 \times 4) \times \sqrt{2 \times 3} = 12\sqrt{6}$$

In general:

$$a\sqrt{b} \times c\sqrt{d} = (ac)\sqrt{bd}$$

करणीभजनम्

Karaṇī-bhajanam (Division of Surds)

भजनं करण्योर्भजनं करण्योर्वगवत् संस्कारः ।
भाज्यकरणीभाजककरणभागे यथोक्तक्रमेण

*bhajanam karaṇyora bhajanam karaṇyora vargavat saṃ
bhājyakaraṇībhājakakaraṇībhāgo yathoktakrameṇa*

:

$$12\sqrt{6} \quad 3\sqrt{2}$$

$$: \frac{12}{3} \times \sqrt{\frac{6}{2}} = 4\sqrt{3}$$

:

$$\frac{a\sqrt{b}}{c\sqrt{d}} = \frac{a}{c} \sqrt{\frac{b}{d}}$$

: **Rule:** In division of surds, the operation is like that for squares. Divide the radical of the dividend by the radical of the divisor, and the coefficient of the dividend by the coefficient of the divisor. **Example:** Divide $12\sqrt{6}$ by $3\sqrt{2}$.

Quotient: $\frac{12}{3} \times \sqrt{\frac{6}{2}} = 4\sqrt{3}$

In general:

$$\frac{a\sqrt{b}}{c\sqrt{d}} = \frac{a}{c} \sqrt{\frac{b}{d}}$$

करणीवर्गः

Karaṇī-vargaḥ (Square of Surds)

वर्गः करण्याः स्यात् पदस्य वर्गः करणी चेति ।
वर्गः करण्याः स्यात् पदस्य वर्गः करणी च भवति

vargaḥ karaṇyāḥ syāt padasya vargaḥ karaṇī ceti
vargaḥ karaṇyāḥ syāt padasya vargaḥ karaṇī ca bhavati

$$: = \times () : 3\sqrt{5}$$

$$: (3)^2 \times 5 = 9 \times 5 = 45$$

$$(3\sqrt{5})^2 = 3^2 \times (\sqrt{5})^2 = 9 \times 5 = 45 :$$

$$(a\sqrt{b})^2 = a^2 \cdot b$$

Rule: The square of a surd equals the square of the coefficient times the radical quantity (the radical sign vanishes). **Example:** Square of $3\sqrt{5}$.

$$\text{Square: } (3)^2 \times 5 = 9 \times 5 = 45$$

Since $(3\sqrt{5})^2 = 3^2 \times (\sqrt{5})^2 = 9 \times 5 = 45$ **In general:**

$$(a\sqrt{b})^2 = a^2 \cdot b$$

करणीमूलम्

Karaṇī-mūlam (Square Root of Surds)

(१) वर्ग करण्य यदि वा करणीस्तु तयो रूपपक्षं वा बहूनाम् ।
विशोध्येष्टं पदं ते शेषस्य रूपाणि युतानि निपातयेत्

(1) *vargaṇi karaṇyā yadi vā karaṇīstu tayo rūpapakṣaṁ vā
bahūnām
viśodhyeṣṭam padaṁ te śeṣasya rūpāṇi yutāni nipātayet*

: $\sqrt{a + \sqrt{b}} = \sqrt{x} + \sqrt{y},$

$$x + y = a$$

$$4xy = b$$

$$\text{atha } x - y = \sqrt{a^2 - b}$$

$$x = \frac{a + \sqrt{a^2 - b}}{2}$$

$$y = \frac{a - \sqrt{a^2 - b}}{2}$$

Rule: To find $\sqrt{a + \sqrt{b}}$, assume $\sqrt{a + \sqrt{b}} = \sqrt{x} + \sqrt{y}$.
Then:

$$x + y = a$$

$$4xy = b$$

Also $x - y = \sqrt{a^2 - b}$. Therefore:

$$x = \frac{a + \sqrt{a^2 - b}}{2}$$

$$y = \frac{a - \sqrt{a^2 - b}}{2}$$

कुट्टकः

Kuṭṭakaḥ (The Pulverizer)

भाज्यं हरेण विभजेद् भाजकेनापवर्त्य तौ ।
यावत्तावत्प्रचलितं कुट्टकं तत्र योजयेत्

*bhājyaṃ hareṇa vibhajed bhājakenāpavartya tau
yāvattāvatpracalitam kuṭṭakam tatra yojayet*

: $ax \pm c = by$, a, b, c ,
 x, y :

1. (a) (b)

2. (: continued fractions)

3. मत्तितः ()

4. : $100x + 90 = 63y$

$$x = 18 + 63t$$

$$y = 30 + 100t$$

Rule: The Pulverizer (*kuṭṭaka*) solves linear indeterminate equations $ax \pm c = by$, where a, b, c are known integers, and x, y are unknown integers. **Method:**

1. Divide dividend (a) by divisor (b) to get quotient and remainder
2. Mutual division (continued fractions)
3. *Matitaḥ* (intelligent) operation
4. Derive general solution from a particular one

Example: $100x + 90 = 63y$

$$x = 18 + 63t$$

$$y = 30 + 100t$$

वर्गप्रकृतिः

Vargaprakṛtiḥ (Square-Nature)

मूलं मूलं द्विगुणं युक्तं तत्प्रकृत्या हतं सहितम् ।
सर्वमूलं पुनर्मूलं ततः प्रकृतिमूलं भवेत्

*mūlaṁ mūlaṁ dviguṇaṁ yuktaṁ tat prakṛtyā hataṁ sahitaṁ
sarvamūlaṁ punarmūlaṁ tataḥ prakṛtimūlaṁ bhavet*

: :

$$Nx^2 + 1 = y^2$$

- :

1. भावना (:):

2. चक्रवाल (:):

$$: 61x^2 + 1 = y^2$$

$$x = 226153980$$

$$y = 1766319049$$

Rule: The Square-Nature (*vargaprakṛti*) deals with:

$$Nx^2 + 1 = y^2$$

known as Brahmagupta's equation or the Pell equation.

Bhāskara's methods:

1. *Bhāvanā* (composition): combining two solutions

2. *Cakravālā* (cyclic method): an iterative algorithm

Example: $61x^2 + 1 = y^2$

$$x = 226153980$$

$$y = 1766319049$$

चक्रवालम्

Cakravālam (The Cyclic Method)

: a, b, k $Na^2 + k = b^2$: m $a \cdot m + b$ k
 , $|m^2 - N|$:

$$a' = \frac{am + b}{k}$$

$$b' = \frac{bm + Na}{k}$$

$$k' = \frac{m^2 - N}{k}$$

: $k' = 1$, (a', b')

The Cyclic Method is Bhāskara's most important contribution to algebra. It always converges to the minimal solution. **Step 1:** Choose a, b, k such that $Na^2 + k = b^2$

Step 2: Find m such that $am + b$ is divisible by k , and $|m^2 - N|$ is minimized. **Step 3:**

$$a' = \frac{am + b}{k}$$

$$b' = \frac{bm + Na}{k}$$

$$k' = \frac{m^2 - N}{k}$$

Step 4: If $k' = 1$, then (a', b') is the solution. Otherwise repeat.

एकवर्णसमीकरणबीजम्

Ekavarṇa-samīkaraṇa-bījam (Seed of Single-Colour Equations)

रूपाणि यत्र स्युः समतुल्यानि तत्र यावत् तुल्यानि ।
यावन्ति तानि तानि विपरीतानि तुल्यानि तु तुल्यानि

*rūpāṇi yatra syuḥ samatulyāni tatra yāvat tulyāni
yāvanti tāni tāni viparītāni tulyāni tu tulyāni*

: संरमण ()

1.

2.

3.

$$: 4yā + 7 = 2yā + 13$$

$$4yā - 2yā = 13 - 7$$

$$2yā = 6$$

$$yā = 3$$

$$: 4(3) + 7 = 19 \quad 2(3) + 13 = 19 \checkmark$$

Rule: In a single-coloured equation, equalize unknowns by transposition (*saṃramaṇa*):

1. Subtract unknown on smaller side from unknown on greater side

2. Subtract known on greater side from known on smaller side

3. Divide the remainder of known by remainder of unknown

Example: $4yā + 7 = 2yā + 13$

$$4yā - 2yā = 13 - 7$$

$$2yā = 6$$

$$yā = 3$$

Check: $4(3) + 7 = 19$ and $2(3) + 13 = 19 \checkmark$

शेषात्समीकरणम्

यद् दातुं यद् दातव्यं तुल्यं तुल्येन संयोजयेत् ।
विपरीतानि तुल्यानि तु तुल्यानि समीकुर्यात्

$$: () 3y\bar{a} + 5 = 2y\bar{a} + 12 \Rightarrow y\bar{a} = 7 ()$$

$$5y\bar{a} - 8 = 3y\bar{a} + 10$$

$$5y\bar{a} - 3y\bar{a} = 10 + 8$$

$$2y\bar{a} = 18 \Rightarrow y\bar{a} = 9$$

$$() 7y\bar{a} + 15 = 4y\bar{a} + 3$$

$$3y\bar{a} = 3 - 15 = -12 \Rightarrow y\bar{a} = -4$$

Śeṣāt-samīkaraṇam (Equation by Residue)

yad dātum yad dātavyam tulyam tulyena samyojayet
viparītāni tulyāni tu tulyāni samīkuryāt

$$\textbf{Examples: (1)} 3y\bar{a} + 5 = 2y\bar{a} + 12 \Rightarrow y\bar{a} = 7 \textbf{(2)}$$

$$5y\bar{a} - 8 = 3y\bar{a} + 10$$

$$5y\bar{a} - 3y\bar{a} = 10 + 8$$

$$2y\bar{a} = 18 \Rightarrow y\bar{a} = 9$$

$$\textbf{(3)} 7y\bar{a} + 15 = 4y\bar{a} + 3$$

$$3y\bar{a} = 3 - 15 = -12 \Rightarrow y\bar{a} = -4$$

असकृत्वर्गादिसमीकरणम्

Asakṛt-vargādi-samīkaraṇam (Repeated Square Equations)

वर्गो घनो वर्गवर्गोऽष्टधातो नवमो वर्गघनश्चेति ।
समीकरणं समं समं कुर्यात् समतुल्यं तु तुल्यम्

*vargo ghano vargavargo 'ṣṭaghāto navamo vargaghanaś ceti
samīkaraṇam samam samam kuryāt samatulyaṁ tu tulyam*

: ()

$$: y\bar{a}^2 - 5y\bar{a} + 6 = 0$$

$$(y\bar{a} - 2)(y\bar{a} - 3) = 0$$

$$\therefore y\bar{a} = 2 \quad y\bar{a} = 3 \quad : 2y\bar{a}^2 + 7y\bar{a} - 15 = 0$$

$$y\bar{a} = \frac{-7 \pm \sqrt{49 + 120}}{4} = \frac{-7 \pm 13}{4}$$

$$\therefore y\bar{a} = \frac{3}{2} \quad y\bar{a} = -5$$

Rule: For equations with squares, cubes, and higher powers, equalize by transposition, then factor or extract roots. (1)

$$\text{Quadratic: } y\bar{a}^2 - 5y\bar{a} + 6 = 0$$

$$(y\bar{a} - 2)(y\bar{a} - 3) = 0$$

$$\therefore y\bar{a} = 2 \text{ or } y\bar{a} = 3 \quad (2) \text{ Quadratic: } 2y\bar{a}^2 + 7y\bar{a} - 15 = 0$$

$$y\bar{a} = \frac{-7 \pm \sqrt{49 + 120}}{4} = \frac{-7 \pm 13}{4}$$

$$\therefore y\bar{a} = \frac{3}{2} \text{ or } y\bar{a} = -5$$

बहुवर्णसमीकरणम्

Bahuvarga-samīkaraṇam (Multi-Coloured Equations)

यावद् यावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः समीकरणे बहुवर्णे प्रयोजयन्ते सततम्

*yāvad yāvat kālaka nīlaka pītaka haritakādayaḥ
varṇāḥ samīkaraṇe bahuvārṇe prayojayante satatam*

:

:

Rule: In multi-coloured equations, multiple unknowns are represented by different colours. **Example:**

$$5x + 3y + 7 = 3x + 5y + 5$$

$$2x - 2y + 2 = 0$$

$$x - y + 1 = 0$$

$$x = y - 1$$

$$5x + 3y + 7 = 3x + 5y + 5$$

$$2x - 2y + 2 = 0$$

$$x - y + 1 = 0$$

$$x = y - 1$$

$$\therefore : y, x = y - 1$$

$\therefore$ Indeterminate solution: for any chosen y , $x = y - 1$

अनेकवर्णमध्यमाहरणम्

Aneka-varṇa-madhyamāharaṇam (Elimination of Many Unknowns)

मध्यमाहरणं तेषां यथा स्थानं प्रयोजयेत् ।
तुल्यानां तुल्यतां नीत्वा मध्यमं हरेद् बुधः

*madhyamāharaṇam teṣāṃ yathā sthānaṃ prayojayet
tulyānām tulyatām nītvā madhyamaṃ hared budhaḥ*

: :

$$2x + 3y + 6 = 0 \quad (1)$$

$$3x + 2y + 7 = 0 \quad (2)$$

$$4x + 5y + 9 = 0 \quad (3)$$

(1) (2) :

$$6x + 9y + 18 = 0$$

$$6x + 4y + 14 = 0$$

$$5y + 4 = 0 \Rightarrow y = -\frac{4}{5}$$

$$: x = -\frac{9}{5}$$

Rule: Elimination of the middle (*madhyamāharaṇa*) is performed by equalizing coefficients of one unknown and subtracting. **Example:**

$$2x + 3y + 6 = 0 \quad (1)$$

$$3x + 2y + 7 = 0 \quad (2)$$

$$4x + 5y + 9 = 0 \quad (3)$$

From (1) and (2):

$$6x + 9y + 18 = 0$$

$$6x + 4y + 14 = 0$$

$$5y + 4 = 0 \Rightarrow y = -\frac{4}{5}$$

Substituting: $x = -\frac{9}{5}$

प्रत्येकसंहारः

Pratyeka-saṁhāraḥ (Individual Subtraction)

प्रत्येकं संहरेद् भाज्यं हरेण प्रत्येकं बुधः ।
लब्धानि यानि तान्येव मूलानि प्रत्येकशः

*pratyekaṁ saṁhared bhājyaṁ hareṇa pratyekaṁ budhaḥ
labdhāni yāni tāny eva mūlāni pratyekaśaḥ*

:
: $100x^2$, $80x^2$, $60x^2$: $20x^2$
: 5, 4, 3 :

Rule: The method of individual subtraction is used when the same divisor is applied separately to multiple dividends. Each quotient is then a separate root. **Example:** Divide $100x^2$, $80x^2$, and $60x^2$ individually by $20x^2$.

Quotients: 5, 4, 3 respectively

प्रदिशविद्यया

Pradiśa-vidyayā (By the Pradiśa Method)

प्रदिशा विद्यया ज्ञेयं मूलं यत्र प्रदिश्यते ।
प्रदिशं तत्र योज्यं स्याद् विद्वद्भिर्नृमीयते

pradiśā vidyayā jñeyam mūlam yatra pradiśyate
pradiśam tatra yojyam syād vidvadbhir anumīyate

: प्रदिश

13824

- 4 4
- $20^3 = 8000$, $30^3 = 27000$
- 20 30
- $\therefore = 24$ ($24^3 = 13824$)

: **Rule:** The *pradiśa* method finds roots by inspection and intelligent guessing, especially useful for higher-degree equations. **Example:** Find the cube root of 13824.

- The number ends in 4, so the cube root ends in 4
- $20^3 = 8000$ and $30^3 = 27000$
- So the root is between 20 and 30
- $\therefore$ Cube root = 24 ($24^3 = 13824$)

नवीनप्रदिशविद्यया

Navīna-pradiśa-vidyayā (By the New Pradiśa Method)

नवीं प्रदिशमास्थाय मूलं यत् प्रदिश्यते ।
तत् सुलभं भवेत् प्राज्ञैर् विदुषां नात्र संशयः

*navīm pradiśam āsthāya mūlaṃ yat pradiśyate
tat sulabhaṃ bhavet prājñair viduṣāṃ nātra saṃayaḥ*

:

Rule: The new *pradiśa* method gives an improved technique for finding roots, suitable for even more complex cases.

This chapter extends the previous *pradiśa* methods with additional techniques for extraction of roots of higher powers and for handling more complex algebraic expressions.

End of Part I / Prathamasya Bhāgasya Samāptiḥ

(Pages 1–57 / Prsthāni 1–57)

Appendix / Paricchedaḥ

संशोधकीयटिप्पणयः

Editorial Notes

. : यावत् (x)
नीलक (z), पीतक (u), हरितक (v)

कालक (y),

1. On notation: Bhāskara uses *yāvat* (x) for the unknown. The colours *kālaka* (y), *nīlaka* (z), *pītaka* (u), *haritaka* (v) represent further unknowns.

. : ऋण ()

2. On negative numbers: Bhāskara fully accepts negative numbers, calling them *ṛṇa* (debt), opposed to *dhana* (wealth/positive). He gives clear rules for arithmetic with negatives.

. : करणी ()
, , , ,

3. On surds: The *karaṇī* (surd) operations demonstrate sophisticated understanding of irrational numbers. Bhāskara can add, subtract, multiply, divide, and extract roots of surd expressions.

. : $Nx^2 + 1 = y^2$ चक्रवाल

4. On the cyclic method: The *cakravālā* method for solving $Nx^2 + 1 = y^2$ is one of the greatest achievements of Indian mathematics. It always converges to the minimal solution.

. :
, , ,

5. On equations: Bhāskara classifies equations into several types and gives systematic methods for each. His treatment of quadratic equations is complete, including cases with two, one, or no real solutions.

Bījagaṇita

The Algebra of Bhāskara II

Part II

Pages 58–114

बीजगणितम् — द्विभाषीसंस्करणम्

□□□□□□□□ □□□□□□□□-□□□□□□□□ □□□□□□□□

अथ एकवर्णसमीकरणम्
अव्यक्तवर्गादिसमीकरणम्
अनेकवर्णसमीकरणम्
अनेकवर्णमध्यमाहरणभेदाः

1. Equations with One Unknown (Ekavarṇa-samīkaraṇa)
2. Equations with Unknown Squares (Avyakta-vargādi-samīkaraṇa)
3. Equations with Many Unknowns (Anekavarṇa-samīkaraṇa)
4. Variations of the Elimination Method (Anekavarṇa-madhyamāharaṇa-bhedāh)

Bhāskara II (1114–1185 CE)

Sanskrit text in Devanagari || English translation with IAST

Contents

[illegible][illegible]

1. $(\mathbb{Z}/N\mathbb{Z})^\times$ 上 χ の $\chi^2 = 1$ となるような χ の組を $\chi_1, \chi_2, \dots, \chi_r$ とし、 $\chi_1, \chi_2, \dots, \chi_r$ は $(\mathbb{Z}/N\mathbb{Z})^\times$ 上で線形独立であるような r を r_1 とする。このとき、 $\chi_1, \chi_2, \dots, \chi_{r_1}$ は $(\mathbb{Z}/N\mathbb{Z})^\times$ 上で線形独立である。

This volume contains **Part II** of the *Bījagaṇita* (“Seed Computation,” i.e., Algebra) composed by **Bhāskara II** (1114–1185 CE). The *Bījagaṇita* is the foundational treatise of Indian algebra, treating equations, unknown quantities, and their solutions. Pages 58–114 encompass four principal chapters: equations with one unknown, equations with unknown squares, equations with many unknowns, and variations of the elimination method.

A distinctive feature of Bhāskara's exposition is that each topic is introduced by a verse (*sūtra*) containing a rule, followed by worked examples (*udāharaṇa*) and detailed commentary (*vārttika*) showing the method of solution step by step. The unknown quantity is denoted by यावत्तावत् (*yāvattāvat*, "as much as"), abbreviated या or याव. Equations are solved by the process of *sama-śodhana* (making equal and clearing).

Key Methods treated in this part include: (1) The **Cakravāla** (cyclic) **method** – Bhāskara’s greatest algorithmic achievement for solving *vargaprakṛti* ($x^2 - Ny^2 = 1$, Pell’s equation); (2) The **Bhāvita** method for product equations; (3) **Madhyamāharaṇa** – the elimination method for equations with multiple unknowns.

Chapter I: Equations with One Unknown

एकवर्णसमीकरणम् || Ekavarṇa-samīkaraṇa

अ[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED], [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED]—[REDACTED] (accepting
side) [REDACTED] [REDACTED] (giving side) [REDACTED] [REDACTED]
[REDACTED] [REDACTED]

Now, equations with one unknown. Here, a single unknown quantity, called **yāvattāvat** (यावत्तावत्, “as much as”), is postulated. Combining it with known quantities (*rūpa*, रूप), one forms an equation and determines the value of the unknown. Each side of the equation is called either the *accepting side* (what is received) or the *giving side* (what is given). By the process of equal-clearing (*sama-śodhana*), the value is determined.

Simple Linear Equation || [REDACTED]

[REDACTED] — [REDACTED] [REDACTED] [REDACTED]
यावत्तावत् [REDACTED] [REDACTED] या १ [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] या ६ [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] या ६ रूप
२० [REDACTED] [REDACTED] या १० [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] या १० रूप १००

Example — Here, the price of a horse is unknown; let its value be one **yāvattāvat**, written **yā 1**. If the price of one horse equals the price of so many goats, then the price in coins is determined. For one horse: **yā 6**. Adding twenty coins (*rūpa*), the first person’s wealth becomes: **yā 6, rūpa 20**. The price of ten horses is **yā 10**. Adding one hundred coins, the second person’s wealth becomes: **yā 10, rūpa 100**.

[REDACTED] — [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED]—

न्यास:---या ६ रूप २०
या १० रूप १००

Equal-clearing — These two are made equal for clearing:

SETTING DOWN: **yā 6, rūpa 20**
yā 10, rūpa 100

[REDACTED] — [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] या ४ [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

Rule — One should clear the unknown from one side, and the known from the other side. When cleared, the remainder on the other side is *rūpa* 400. Then, dividing *rūpa* 400 by the coefficient of the unknown (*yā* 4), the value of one *yāvattāvat* is obtained as 100. Thus the price of one horse is 100 (coins).

Example: Jewels and Wealth /

माणिक्क्यामल्लीलमौक्तिकमितिः पञ्चाषसममा—
देकस्यान्यतरस्य सप्त नव पट् तद्दलसंख्या सखे।
रूपाणां नवतिद्विषष्ठिर्नयोस्तौ तुल्यवित्तौ तथा
वीजं प्रतिरुज्जानि कुरुते मौल्यानि शीघ्रं वद ॥३॥

एको द्वीति मम द्विहि शतं धनेन
त्वत्तो भवामि हि सखे द्विगुणस्ततोऽस्यः।
भूते द्वयोर्दसि वै नम बहुगुणोऽहं
त्वत्तस्तयोर्वद धने मम किं प्रमाणे ॥४॥

Verse 3: Jewels and Equal Wealth

“The measures of ruby, sapphire, pearl, and emerald of one are fifty, seven, nine, and half that number, O friend. Their total wealth is one hundred sixty-two coins each. Tell me quickly the prices, O algebraist, that make them equal.”

SOLUTION: Let the prices of ruby etc. be $yā(1)$ 3, $yā$ 2, $yā$ 1. The two sides become:

**$yā$ 19, $yā$ 16, $yā$ 7, $rūpa$ 90
 $yā$ 21, $yā$ 18, $yā$ 6, $rūpa$ 62**

By equal-clearing, the value of $yāvattāvat$ is 4. The prices of jewels are 12, 8, 4.

Verse 4: The Merchant's Wealth

“If you give me one hundred, O friend, I become twice as rich as you. But if I give you two hundred, I become some multiple. Tell me, what are our respective measures of wealth?”

SOLUTION: Let the two merchants' wealth be $yā$ 1 and $yā$ 1. After giving 100, the first becomes twice the second:

$yā$ 1, $rūpa$ 100 = $yā$ 2, $rūpa$ 200. By clearing: **$yā$ 4, $rūpa$ 300**, whence $yā = 75$. The wealths are 40 and 170.

The diagram illustrates the decomposition of a 10x10 grid into a 10x5 grid and a 5x5 grid. The 10x10 grid is shown as a large rectangle divided into four 5x5 quadrants. The 10x5 grid is shown as a rectangle divided into two 5x5 quadrants. The 5x5 grid is shown as a single 5x5 quadrant. The decomposition is represented by the equation: 10x10 = 10x5 + 5x5.

याव ३ या ३१ रूप ७५
याव ० या ० रूप १६९

Equating and multiplying both sides by twelve, taking the double-root from the square 169, the roots are **yā 6**, **rūpa 31** and **rūpa 43**.

The Lotus and the Swallow / ॥८॥

॥८॥ — ॥८॥ ॥८॥ ॥८॥
 ॥८॥ ॥८॥ ॥८॥ ॥८॥
 ॥८॥ ॥८॥ ॥८॥ ॥८॥ ॥८॥
 ॥८॥ ॥८॥ ॥८॥

Example: The Lotus and the Swallow — Here, the difference of the two root-measures, multiplied by the product of the multiplier, equals the second root-wealth. The multiplier of the second is 2. If the first root-wealth equals the product of one multiplier and the second root-wealth, divided by the square of the difference, then the difference of measures is 4. Thus the first and second root-wealths are 8 and 4, yielding a result of 2.

एकक्रान्तर्ययनात् फलस्य वर्गं विशोध्य परिशिष्यम्।
 पञ्चक्रान्तेन दत्तं तुल्यः कालः फले च तथैः ॥८॥

Verse 8: The Lotus Problem

“Having subtracted the square of the result from the first increased [quantity], the remainder, divided by five, gives equal time and result.”

SOLUTION: The multiplier is 5. Dividing by the decreased multiplier (4), the square of the small difference (16) divides the second wealth, yielding 4. Adding the square of the difference gives the first wealth as 20. By the rule of proportion, time = 20.

Four Merchants with Equal Wealth / ॥॥॥॥॥॥॥ ॥॥॥॥ ॥॥॥॥॥॥॥

म०णिक्पाणुकणिनीलादीनां मुक्ताफलानां शतं
षड्भागि च पञ्च रत्नवीरां येषां चतुर्णां धनम्।
संग्रहे हयरेन ते निजधनाद्वेकैक मिथो
जातास्तुल्यधनाः पृथग्वद सखे तद्नमैल्यानि मे ॥२॥

Verse 2: Four Merchants of Equal Wealth

“Of rubies, coral, sapphires, pearls, and emeralds, there are one hundred; six parts and five jewels. Four merchants whose wealth, when one gives to another, they become equal. Tell me separately, O friend, the prices of their wealth.”

METHOD: The unknowns are postulated as yāvattāvats. After four equal-clearings, the root is obtained by intelligent reasoning. As Bhāskara himself says:

“Neither squared nor unsquared is the seed; seeds are not separate. Intelligence alone is the seed, for endless are the suppositions.”

Geometric Application: Equilateral Triangle / XXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXX

XXXXXX — XXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXX XXXX
XXXXXXXXXX = याव १ रूप १५, रूप २०० XX X XX X X XXXX
XX XXXXXXXXXXXXXXXXXXXX = याव १ या क ६२ या २ रूप १९ क ६२

XXXXXXXXXX — XXXX XXXXXXXXXX XXXXXXXXXX XXXX XXXX XXXXXX—

र २५ क ५१२
या २ या क ६२

Setting Down: From geometric computation for an equilateral triangle, the square of the altitude is: **yāv 1, rūpa 15, rūpa 200**, or **yā 1, kā 18, rūpa 1**. The second altitude-square is: **yāv 1, yā kā 62, yā 2, rūpa 19, kā 62**. Subtracting the base 6, the second altitude-square becomes: **yāv 1, yā 2, yā kā 62, rūpa 13, kā 62**.

Equal-clearing: These being made equal, the sides after clearing are:

rūpa 25, kā 512
yā 2, yā kā 62

Equations with Unknown Squares /

 —

अव्यक्तवर्गादि यदा द्वयोः पक्षौ तदैकेन निःशेषं किञ्चित्।
क्षेपं तयोरेन पदप्रदः स्याद्व्यक्तः सहेत परेन भूयः ॥१॥

व्यक्तस्य मूलस्य समीकृते समव्यक्तमानं बहु लभ्यते
तत्।
न निर्वहेत् चेद्धनवर्गयोस्त्वेति तदा क्षेपमिदं स्यबुद्ध्या ॥२॥

अव्यक्तमूलकां ततोऽस्य व्यक्तस्य पक्षस्य पदं यदि
स्यात्।
मूलं धनं तथा विधाय साव्यमव्यक्तमानं द्विधा कल्पितं
स्यात् ॥३॥

Rule (Verses 1–2): —

*When unknown squares appear on both sides,
divide by something to make one side a perfect square.*

*The additive must be such that it yields
a rational root when combined with the known term.*

*When the root of the known is equated,
many unknown values may be obtained;
if the square does not succeed,
then by intelligence determine the additive.*

Verse 3:

*If the root of the unknown side, having a number,
when made the root of the known side—
having made that root positive or negative,
the unknown value is to be assumed twofold.*

Example: Series of Numbers /

स्वार्धपञ्चाशनवमैर्युक्ताः के स्युः समाश्रयाः।
अन्यैर्द्वादशहीनाख्य परिशिष्टाख्य तान् वद ॥१४॥

Verse 14: Numbers with Fractional Relations

“What numbers, when increased by their own half, five-ninths, and [other fractions], become equal? And when diminished by others, twelve less—tell me the remainders.”

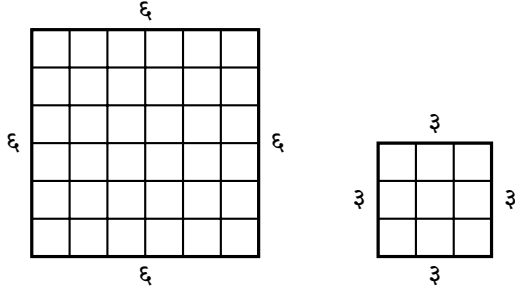
SOLUTION: Let the equal number be yāvattāvat 1. By the reverse process, the numbers are $yā \frac{2}{5}$, $yā \frac{5}{6}$, $yā \frac{9}{10}$. Dividing by the greatest common measure and multiplying by yāvattāvat value 150, the numbers are 100, 125, 135.

त्रयोदश तथा पञ्च करण्यो मूजयोनितो।
भूजाता च चत्वारः फले भूमिं वदाऽखु मे ॥१५॥

Verse 15: The Excavated Plot

“The square-roots of thirteen and five, diminished by the base, and four as the result—tell me the area, O algebraist.”

SOLUTION: Let the area be yāvattāvat 1. By the formula “altitude times half-base divided [equals the area of a triangle],” the section-ratio is $kā \frac{5}{13}$. Its square is $kā 5$. Subtracting from rūpa 5, rūpa $\frac{5}{13}$, the root gives the altitude $kā \frac{5}{13}$.

Geometric Diagrams /  —   **Geometric Figures:** Their forms are as follows:

Square of side 6 Square of side 3

The large square is 36 units in area; the small square is 9 units. The difference is 27, which may be analyzed as the sum of gnomon regions.

चतुर्गुणस्य
घातस्य
युतिवर्गस्य
चान्तरम्।
राश्यन्तरकृतेस्तुल्यं
द्वयोरन्त्यकयोर्थथा
॥१७॥

Verse 17:

The difference between the
square of the sum
and four times the product
equals the square of the
difference of the two
numbers,
as between the last two
terms.

NUMERICAL VERIFICATION:

For numbers 3 and 5:

$$(3 + 5)^2 - 4 \cdot 3 \cdot 5 =$$

$$64 - 60 = 4 = (5 - 3)^2.$$

This is the algebraic
identity:

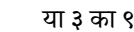
$$(a + b)^2 - 4ab = (a - b)^2.$$

Chapter II: Equations with Many Unknowns

अनेकवर्णसमीकरणम् || Anekavarṇa-
samīkaraṇa

अ      

Now, equations with many unknowns. Here, equations are given involving several associated yāvattāvats. When the values of many unknown quantities must be determined, a yāvattāvat is postulated for each quantity. Thereafter, the values are determined by the *kutṭaka* (pulverizer) method.

 —           

न्यासः—या ५ का ८ नी ३ रूप १०
या ३ का ९ नी ५ रूप ६२

Main Rule: Let the prices of rubies etc. be postulated as yāvattāvats etc., and multiplying by their respective coefficients and adding rūpas, for equal-clearing:

SETTING DOWN: yā 5, kā
8, nī 3, rūpa 10
yā 3, kā 9, nī 5, rūpa 62

Equal-clearing: Clearing

$x^2 - Ny^2 = 1$
 को हल करने के लिए, हम N के गुणकों (factors), x के गुणकों (divisors), y के गुणकों (divisors) को
 विचार करते हैं।

1. $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$ $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$
2. $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$ $\mathbb{Z}[\sqrt{N}]$: $a^2 - N \cdot 1^2 = k$, $\mathbb{Z}[\sqrt{N}] a$
 $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$, $k \in \mathbb{Z}[\sqrt{N}]$
3. $b = \frac{a+m}{|k|} \in \mathbb{Z}[\sqrt{N}]$, $m \in \mathbb{Z}[\sqrt{N}]$
 $\mathbb{Z}[\sqrt{N}] a + m \equiv 0 \pmod{|k|} \iff m^2 - N$
 $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$
4. $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$: $a_{\text{new}}^2 - N \cdot b^2 = k_{\text{new}} \in \mathbb{Z}$
5. $k_{\text{new}} = 1, 2, -1, -2, 4, -4 \in \mathbb{Z}$, $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$
(composition) $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$
6. $\mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}] \rightarrow \mathbb{Z}[\sqrt{N}]$

1. Take the integer nearest to $\sqrt{N}$.
2. Form the first equation: $a^2 - N \cdot 1^2 = k$, where a is the nearest integer, k is the additive (*kṣepa*).
3. Compute $b = \frac{a+m}{|k|}$, where m is chosen such that $a+m \equiv 0 \pmod{|k|}$ and $m^2 - N$ is minimized.
4. Form the new equation: $a_{\text{new}}^2 - N \cdot b^2 = k_{\text{new}}$.
5. When $k_{\text{new}} = 1, 2, -1, -2, 4, -4$, apply *bhāvanā* (composition).
6. Otherwise, repeat the cycle.

The Famous Example $N = 61$ / $N = 61$

— एकषष्टिप्रकृतिः'' $N = 61$ $x^2 - 61y^2 = 1$

Example: The “*prakṛti* sixty-one” is Bhāskara’s most celebrated example. Here $N = 61$. The minimal solution of $x^2 - 61y^2 = 1$ is enormously large, and was considered the most difficult case from the time of Brahmagupta until Bhāskara.

—

चक्रम्	गुणकः	ज्येष्ठम्	कनिष्ठम्	क्षेपः	वृत्तिः
१	६१	८	१	-१३	प्रारम्भः
२	६१	४७	६	७	चक्रम्
३	६१	४८	७	-१३	चक्रम्
४	६१	६३	८	-४	भावना
५	६१	११९	१५	४	भावना
६	६१	३९७	५१	-४	भावना
७	६१	१३२३	१७०	७	चक्रम्
८	६१	१३२४	१७१	-१३	चक्रम्
९	६१	१८९१	२४२	७	चक्रम्
१०	६१	२८५६	३६५	-३	चक्रम्
११	६१	२८५७	३६६	-१	समाप्तिः

Cycles of the Cakravālā:

Cycle	Prakṛti N	Jyeṣṭha a	Kaniṣṭha b	Kṣepa k	Operation
1	61	8	1	-13	Init
2	61	47	6	7	C
3	61	48	7	-13	C
4	61	63	8	-4	Bh
5	61	119	15	4	Bh
6	61	397	51	-4	Bh
7	61	1323	170	7	C
8	61	1324	171	-13	C
9	61	1891	242	7	C
10	61	2856	365	-3	C
11	61	2857	366	-1	Term

HISTORICAL NOTE: This solution was not matched in Europe until the work of Fermat, Brouncker, and Wallis in the 17th century. Bhāskara’s Cakravāla algorithm always terminates—a fact that was only rigorously proved in the 20th century by Krishnaswami Alladi. The method is superior to the European “chain method” of Brouncker in both elegance and efficiency.

The Bhāvita Method ||

भविता — भावित (product) (composition)
 (composition) (composition)

— (a, b, k) (A, B, K)
 $a^2 - Nb^2 = k$ $A^2 - NB^2 = K$, then by *bhāvanā* a new solution is obtained:

$$x = aA \pm NbB, \quad y = aB \pm Ab$$

$x^2 - Ny^2 = kK$

The Bhāvita Method: *Bhāvita* means the product of two unknown quantities. When the product of two unknowns appears in an equation, the *bhāvita* method is applied. The principal application of this method is in the *bhāvanā* (composition) operation for solving the *vargaprakṛti*.

The Bhāvanā Rule: If (a, b, k) and (A, B, K) are two solutions of the *vargaprakṛti*, i.e., $a^2 - Nb^2 = k$ and $A^2 - NB^2 = K$, then by *bhāvanā* a new solution is obtained:

$$x = aA \pm NbB, \quad y = aB \pm Ab$$

whose additive satisfies the new equation $x^2 - Ny^2 = kK$.

SPECIAL CASE: When $k = K = 1$, the product $kK = 1$, so the composed solution directly solves $x^2 - Ny^2 = 1$. When $k = K = -1$, then $kK = 1$ again, so two solutions with additive -1 compose to a solution with additive $+1$. This is the key to terminating the Cakravālā method.

:

- $$\begin{array}{l} \text{XXXXXXXXXXXXXXXXXXXX} - \text{XXXXXXXXXXXXXXXXXXXX} \\ \text{XXXXXXXXXXXXXXXXXXXX} \times \text{XXXXXXXXXXXX} \end{array}$$
- $$\begin{array}{l} \text{XXXXXXXXXXXXXXXXXXXX} - \\ \text{XXXXXXXXXXXXXXXXXXXX} \times \text{XXXXXXXXXXXX} \end{array}$$
- $$\begin{array}{l} \text{XXXXXXXXXXXXXXXXXXXX} - \text{XXXXXXXXXXXXXXXXXXXX} \\ \text{XXXXXXXXXXXXXXXXXXXX} \times \text{XXXXXXXXXXXX} \end{array}$$
- $$\text{XXXXXXXXXXXX} - \text{XXXXXXXXXXXX} \text{XXXXXXXXXXXXXXXXXXXX}$$
- $$\begin{array}{l} \text{XXXXXXXXXXXXXXXXXXXX} - \text{XXXXXXXXXXXXXXXXXXXX} (x^2 - Ny^2 = 1) \\ \text{XXXXXXXXXXXX} \text{XXXXXXXXXXXXXXXXXXXX} \end{array}$$
- $$\begin{array}{l} \text{XXXXXXXXXXXX} - \text{XXXXXXXXXXXXXXXXXXXX} \text{XXXX} \\ \text{XXXXXXXXXXXX} \times \text{XXXXXXXXXXXX} \end{array}$$

- $\text{minimize } \sum_{i=1}^n x_i$ (always terminates)
- $\text{minimize } \sum_{i=1}^n x_i$ (produces the minimal solution)
- $\text{minimize } \sum_{i=1}^n x_i$ (more efficient than European methods)
- $\text{minimize } \sum_{i=1}^n x_i$ (foundational for modern algorithmic analysis)

1. **Equations with One Unknown** – solution of linear and quadratic equations.
2. **Equations with Unknown Squares** – solution of equations containing unknown squares.
3. **Equations with Many Unknowns** – solution by the *kuttaka* (pulverizer) method.
4. **Elimination of the Middle** – the method of eliminating the middle unknown.
5. **The Cakravāla Method** – the supreme method for solving $x^2 - Ny^2 = 1$.
6. **The Bhāvita Method** – product equations and the composition operation.

Bhāskara's Contribution: Bhāskara's **Cakravāla method** is one of the most significant achievements in the history of mathematics. This method:

- Always terminates (always terminates)
- Produces the minimal solution (produces the minimal solution)
- Is more efficient than European methods of the 17th century (more efficient than European methods)
- Is foundational for modern algorithmic analysis (foundational for modern algorithmic analysis)

Bījagaṇita

Algebra of Bhāskara II

Part III: Pages 115–170

BILINGUAL ENGLISH–SANSKRIT EDITION

(Parts 7–9 of 9)

श्रीभास्कराचार्यविरचितं बीजगणितम्

Śrībhāskarācāryaviracitaṃ Bījagaṇitam

The Bījagaṇita (Algebra) composed by Śrī Bhāskarācārya

Parallel-Column Layout:

LEFT: Sanskrit original in Devanagari script

RIGHT: English translation with IAST transliteration of technical terms

Covering advanced topics in *Varga-prakṛti* (Pell's equation),
Aneka-varṇa-madhyamāharaṇa-bhedāḥ (equations with multiple unknowns),
Bhāvitam (product equations), and concluding verses

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Introduction

This volume presents the final portion of Bhāskara II's *Bijaganita*, comprising pages 115–170 of the original Chowkhamba Sanskrit Series edition. This section covers the most advanced topics in the treatise:

1. **Aneka-varṇa-madhyamāharaṇa-bhedāḥ** (Equations with Multiple Unknowns) — systematic methods for solving equations involving several unknown quantities, building on the single-variable equations treated earlier.
2. **Bhāvitam** (Product Equations) — equations involving products of unknowns, including the famous “Bhāvita” (product) class of problems.
3. **Śeṣaṇa and Prakṣepaka-vidhi** — special algebraic methods for remainder problems and the method of addition/substitution.
4. **Concluding Verses** — the epilogue (*upasaṃhāra*) in which Bhāskara summarizes his work and pays homage to tradition.

Bhāskara employs his characteristic notation where unknown quantities are denoted by the initial syllable याव (*yāva-*, from *yāvattāvat*, “as much as”), and coefficients are prefixed to these symbols. The final sections also include a publisher's catalogue listing the many Sanskrit texts available from the Chowkhamba Sanskrit Book Depot, Varanasi.

1 From the Varga-prakṛti Section (p. 115)

The opening of this portion continues the discussion of the Varga-prakṛti (square-nature) equation, i.e., equations of the form $Nx^2 + k = y^2$. The text presents further rules and examples for finding solutions to such indeterminate equations.

Continuation of the Square-nature Method

समशोधने हृते पक्षौ यावच्चावत्तावकस्य रूपं प्रक्षिप्य प्रथमपक्षमूलं या २ रू १ ।
परपक्षस्य काव १२ रू १ । वर्गप्रकृत्या मूले क २ उचे ७ वा क २८ उचे ९७ ।
कनिष्ठं कालकमानम् । उच्चिष्टस्य या २ रू १ समं हत्वा लब्धं यावच्चावकमानम् ३ वा ४८ ।
स्वच्छानेनोत्थापने हृते जाती राशी १, ५ वा २०, ७६ इत्यादि ॥

*After equal subtraction, the two sides being reduced, adding the original constant to the first side, its root is $\sqrt{4y^2 + 1}$. For the second side: $\sqrt{12y^2 - 1}$. By the square-nature [method], the roots are (2, 7) or (28, 97). The lesser is the time-value. The residue from $\sqrt{4y^2 + 1}$ gives the *yāvattāvat* value 3 or 48. By self-substitution and reduction, the series of values 1, 5 or 20, 76, etc.*

अथापरं सूत्रं साध्येऽस्तु ॥

(१) द्वितीयपक्षे सति सममेव तु हत्यापवर्त्योर्न पदे प्रसाध्ये ।
उच्चिष्टं कनिष्ठेन तदा निहत्याच्छेदं वर्गेण हतोऽपवर्तः ॥ ६ ॥
कनिष्ठवर्गेण तदा निहत्योच्चिष्टं ततः पूर्वदेव शेषम् ।
इत्यर्थात् ॥

Rule 6: “When the second side exists [and is not unity], having made [the two sides] equal, multiplying and reducing, the remainder [is obtained]; having multiplied the residue by the lesser [root], dividing by the square [nature], the reduced [equation is obtained]. Then having multiplied by the square of the lesser [root], the residue — from that, as before, the remainder.” Thus the meaning.

Worked Example (p. 115)

(१) वि०—कल्प्यते समी पक्षौ

काव १

यावव्। ई १ याव।ई १

अत्र प्रथमपक्षस्य मूले क १ द्वितीयपक्षस्य यावव्।ई १ याव।ई १ मूलेन सममिति ।
तत्र द्वितीयपक्षस्य मूले च का।

$$\boxed{\text{का}} = \sqrt{\boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}}} = \boxed{\text{का}} \sqrt{\boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}} \boxed{\text{का}}}$$

अत इदं याव।ई १ ई १ मूलदं ततो वर्गप्रकृतिविषयो यथा का वर्गः ई, गुणः ई, युतो मूलदं इति हत्वा यावच्चावकमानं उच्चिष्टं चास्य याव।ई १ ई १ मूलेन सममिति।

Summary Rule (p. 115 bottom)

The section concludes with a general procedure:

एवं बहुधा बुद्धिमत्त्वविचिन्त्यामिति सर्वमुपपन्नम् ॥

“Thus in many ways, by the thoughtful application of intelligence, all [is demonstrated].” Everything is proved.

2 Aneka-varṇa-madhyamāharaṇa-bhedāḥ (pp. 116–133)

This is the largest section of the final portion, dealing systematically with equations involving multiple unknown quantities (*aneka-varṇa*, “many colours”). The Sanskrit term *varṇa* (colour) is the traditional designation for an unknown quantity in Indian algebra — different unknowns being assigned different colours.

2.1 General Rules for Multiple Unknowns (pp. 116–117)

अथ सूत्रं साध्येऽस्तु ॥

(१) द्वितीयपक्षे सति सममेव तु हत्यापवर्त्योन पदे प्रसाध्ये ।

उच्चिष्टं कनिष्ठेन तदा निहत्याच्छेदं वर्गेण हतोऽपवर्तः ॥ ६ ॥

कनिष्ठवर्गेण तदा निहत्योच्चिष्टं ततः पूर्ववदेव शेषम् ।

इत्यर्थात् ॥

Worked Example 1 (p. 117)

उदाहरणम् ॥

यस्य वर्गेऽद्वितीः पञ्चगुणा वर्गशतोनिता ।

मूलद्वा जायते राशिः गणितज्ञ वदाशु तम् ॥ १ ॥

Example 1: “A number whose square minus twice [itself], diminished by a hundred times five [i.e., 500], has two roots — tell me that number quickly, O mathematician.”

अत्र राशिः=या १ । अस्य वर्गेऽद्वितीः पञ्चगुणा वर्गशतोनिता यावव ५ याव १०० । अयं वर्ग इति कालकवर्गसमं हत्वा गृहीतं कालकवर्गस्य मूलं का १ । द्वितीयपक्षस्य यावव ५ याव १०० ।

यावच्चावकगुणापवर्त्य वर्गप्रकृत्या मूले क १० उचे २० वा क १७० उचे ३८० ।

“Having assumed the root of twice the divisor of the first side as the ‘blue’ [unknown], having divided the other side by its square, the desire is satisfied more than previously stated.”

2.2 The Method of Substitution (pp. 125–128)

अथ सूत्रं सूचयम् ॥

(१) सकृपमयकमल्पकं वा विभोगमूलं प्रथमं प्रकल्प्य ।

योगान्तरशेषकमाजिताच्छेदान्तरशेषकतः पदं स्यात् ॥ १२ ॥

Rule 12: “Having assumed the root of the difference of [the two sides] as the first [unknown], or a small quantity, [and] the residue of the difference of sums as the square [of another unknown], the root [is obtained].”

(१) वि०—अत्र कल्प्यते योगान्तरशेषमानम् = क्षे

वर्गान्तरशेषमानम् = क्षे₂, वर्गयोगशेषमानम् = क्षे₃

विभोगमूलम् = या, योगमूलम् = का

$$\text{क्षे}^2 - \text{क्षे}_1 = \text{क्षे}^2 - \text{क्षे}_1, \text{क्षे} = \text{क्षे}^2 - \text{क्षे}_1$$

$$\text{क्षे} = \frac{\text{क्षे}^2 - \text{क्षे}_1}{2}, \text{क्षे} = \frac{\text{क्षे}^2 + \text{क्षे}_1 - 2\text{क्षे}_1}{2}$$

Rule 13 (p. 129)

शेषं ततः शेषकमूलमन्यच्च मूले विद्ध्वादसकृत् सम्मते ॥ ९ ॥

समाविते वर्गकृती तु यत्र तन्मूलमादाय च शेषकस्य ।

इष्टोद्धृतस्येर्विवर्जितस्य दलैन समं तद्विह तदैव कार्यम् ॥ १० ॥

Worked Example on the Product Method (p. 130)

उदाहरणम् ॥

तौ राशी वद यत्कृत्योः समस्तगुणयोर्युतिः ।

मूलद्वया स्याद्योगमष्टौ मूलद्वय रूपसंयुतः ॥ १ ॥

अत्र राशी या १, का १ । अनयोर्वर्गयोः समस्तगुणयोर्युतिः यावव ७ काव ८ । अयं वर्ग इति नीलकवर्गसमं हत्वा पक्षयोः कालकधनं प्रक्षिप्य नीलकपक्षस्य मूलं नी १ । परपक्षस्य याव ७ काव ८ ।

Summary Verses (p. 131)

अथ सूत्रं सूचयम् ॥

(१) सकृपमयकमल्पकं वा विभोगमूलं प्रथमं प्रकल्प्य ।

योगान्तरशेषकमाजिताच्छेदान्तरशेषकतः पदं स्यात् ॥ १२ ॥

वर्गान्तरशेषकसमितिर्युता क्षेपेण हत्वोर्ध्वतेन तै ततः ।

द्विधा पदं तत्पदयुगविभोगाज् मूलं युतमूलमतस्तयोरिति ॥

“The difference of squares having a residue equal to the divisor, divided by the additive, then by the quotient, [the root] divided in two; from the difference of the pair of roots, the root [of the lesser] from the sum of the roots, thus [proceeding]...”

3 Bhāvitam and Prakṣepaka-vidhi (pp. 134–152)

3.1 Equations with Product Terms (pp. 134–138)

This section deals with *bhāvita* (product) equations, where the unknowns appear in product form. The method extends the techniques for multiple unknowns to handle more complex algebraic structures.

अथ वा स्वर्णकल्पनायां मन्दबुद्धीनां पुनः किञ्चित् पठितः ।
तत् सूचयामि ॥

हरमका यस्य हति सूक्ष्मति सोऽपि द्वितीयपदयुतातः ।
तेनाहतोऽस्यवर्णो रूपपदेनान्वितः कल्पः ॥ १६ ॥

Rule 16: “The divisor multiplied by [some quantity] whose [result] is fine, added to the second term; having multiplied by that, the unknown [is obtained], combined with a root term [as] the assumption.”

Worked Example (p. 137)

उदाहरणम् ॥
न यदि पदं रूपाणां शिष्येत्तं तेन हरतोऽपि ।
नावच्छद्गुणो भवति न चेदेवमपि किल तथै ॥ १७ ॥

हरमतिः । यस्याङ्कस्य हतिहरमका सती सूक्ष्मतीति निश्शेषा भवति अपि च सोऽप्यङ्को हर्यो रूपपदेन च गुणितो हरमकः सन् सूक्ष्मति तदा तेनाहतोऽस्यवर्णस्तेन रूपाणां घनमूलं न लभ्यते तदा तेन रूपेषु हरतोऽपि तावद्गुणो भवेत् तन्मूलं रूपपदं स्यात् ।

3.2 The Bhāvita Rule (pp. 140–141)

भावितम् ॥
वर्णद्वयातिकूपेऽस्य मकरवेष्टनैकतः ।
एताम्या संयुताबुद्धि कल्पयो वैच्छया च तौ ॥ ३ ॥
वर्णद्वयो वर्गयोरन्ते क्रियते ते विपर्ययात् ।
समयोः पक्षयोरेकमाध्यमिकमपास्य तौ वर्णौ रूपाणि च

The section on “Product”: “Two unknowns, their product having a coefficient, from one side having a single enclosure [i.e., bracketed], having assumed any sum of their values, [proceeding] with the pair of unknowns reversed, [taking] the square of two unknowns, the means [being] equal, having removed one middle term, the two unknowns and the roots...”

Worked Example on Bhāvita (p. 142)

उदाहरणम् ॥
चतुःषष्टिगुणयो राश्योः संयुतिर्द्वियुता तयोः ।
राशिघातेन तुल्येति ॥

तत्र यथोक्तं हते पक्षौ { या ४ का ३ रू २
या।का।भा १

वर्णद्वयातिकूपेऽस्य १४ एतदेकेनाहतं हते जाते द्विके १, १४ । एते वर्णद्वया ४, ३ स्वैच्छया युते जाते यावच्चावककालकमाने ५, १८ वा १७, ५ । द्विकेन ५, ११ वा १०, ६ ॥

3.3 The Method of Mean Elimination (pp. 145–147)

माध्यमाविगमः ॥
आसन्नमानस्य हराशमानम् अप्राप्तिगुण्ये सहिते क्षेपेण ।
पूर्णे स्याताम् अन्तरराशिकाभ्यां तदा हरांशो भवतीश्रमस्य ॥ १ ॥
आसन्नमानयोरन्तरमेव भवेत् ।
अंशस्थाने सदा रूपं चिन्त्येत तथा धीमता ॥ २ ॥
सर्वेषामासन्नमानेषु हरांशो भवति षट् ।
तथोत्तरोत्तरं सूक्ष्मतामासन्नानि भवन्ति हि ॥ ३ ॥

The method of mean-elimination: “The divisor-measure of the approximate value, added to the non-obtained quantity and the additive, being complete by two numbers with difference, then the divisor-fraction [gives] the effort. The difference of two approximate values alone [suffices]. The numerator [is] always unity [at first], thus by the intelligent. For all approximate values, the divisor-fraction is six. Thus gradually, more and more subtle approximate values are obtained.”

Concluding Verses of the Bhāvitam Section (pp. 151–152)

तथा गोले मयोक्तम् ।
 उत्थ सदसतां मर्तानां वैराशिकमात्रमेव पाठो बुद्धिरेव बीजम् ।
 तथा गोलाख्याय मयोक्तम् ।
 अस्ति वैराशिकं पाठी बीजं च विमला मतिः ।
 किमन्यत् सुखदीनामतो मन्दाऽर्थमुच्यते ॥
 गणककर्मान्तरमयं बाललोचनगम्यं
 सकलगणितसारं सोपपट्टिप्रकारम् ।
 इति बहुगुणयुक्तं सर्वदोषैर्विमुक्तं
 पठ पठ मतिबुद्धयै लाघवं प्रौढिसिद्धयै ॥

The epilogue: “Thus I have spoken of the sphere. For people ascending from the unreal to the real, the mere reading [of this text] is the seed of intelligence. Thus I have spoken of the sphere called ‘Gola’. There is the study of the text and pure intellect — what else [is needed]? For the delight of the wise, this is declared. This work accessible to the eye of a child [yet] profound, the essence of all mathematics with proofs endowed, endowed with many virtues, free from all faults — read, read with intelligence and understanding, for ease and the attainment of maturity.”

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
 बीजगणिताध्यायः समाप्तः ॥

“Thus ends the Bījagaṇita-adhyāya in the Siddhāntaśiromaṇi composed by Śrī Bhāskarācārya.”

वि०—इति कूपाङ्कनमुखाकरे यदिह भास्करबीजमपूर्वकम् ।
 तदुपनिषमतीव चमत्कृतिं विभिधरस्य चकार च कारणम् ॥

Editor’s note: “That unprecedented algebra of Bhāskara which here [appears] with the face of a well-dig-mine [i.e., a numerical source], having ascertained it with the Upaniṣad [of commentary], the cause of the appearance of manifold wonder.”

4 Final Sections and Publisher’s Catalogue (pp. 153–170)

4.1 Navīna-mati-viśeṣaḥ (pp. 153–157)

The section on “Special Methods for New Understanding” presents advanced techniques for solving equations through approximation and iteration, building on the methods developed earlier.

नवीनमतिविशेषः ॥
 निरक्ष पदं यद्विगुणात् स्यात् फलार्थं धनार्थं तदेवात्र शेषं तदस्यम् ।
 पदद्वयं धनं शेषहतप्रमन्यत् फलं तदभवत् शेषमूलं धनेन ॥ १ ॥
 धनार्थं नवं तस्य कृत्या विहीनो गुणः शेषमकोऽस्यवर्णस्य मानम् ।
 मूदगुसेव मते यदा शेषमानं भवेद्वितीयं तदा लब्धितौ ते ॥ २ ॥

Rule 1: “A number, when doubled, becomes the root; for the fruit [result], for wealth [coefficient], that same residue here — its two terms [are] the wealth divided by the residue; the fruit [is] the root of the residue multiplied by the wealth. The new wealth is diminished by nine times its square; the multiplier, diminished [thus], [gives] the measure of the unknown. When by the opinion of Mūḍgu [i.e., by the method of approximation], the residue-value becomes the second [term], then the quotient for both.”

4.2 Worked Examples on Special Methods (pp. 157–159)

उदाहरणानि ॥

(१) कल्प्यते $अ+क=५७६०$, $अ-क=\frac{अ}{१०}$ तदा $अ=३८४०$,
 $क=१९२०$ इति क्षिप्रम् ॥

(२) यदि $\frac{क+१०}{१०} = \frac{क+१०}{१०}$ तदा कमानम्=१ इति क्षिप्रम् ॥

(३) यदि $\frac{क+१०}{१०} + \frac{क-१०}{१०} + \frac{१०}{१०} = \frac{क+१०}{१०}$, तदा $क=२०$ इति क्षिप्रम् ॥

(४) $(क + \frac{क}{१०})(क - \frac{क}{१०}) - (क + १०)(क - १०) + \frac{क}{१०} = १०$
तदा $क=१२$ इति क्षिप्रम् ॥

4.3 The Method of Approximate Roots (pp. 160–165)

This section presents the *āsanna-mūla* (approximate root) method — an iterative technique for finding successively better approximations to irrational square roots, a precursor to what is now known as Newton’s method.

आसन्नमूलानि सूत्राणि ॥

आसन्नमानस्य हराशमानम् अप्राप्तिगुण्ये सहिते क्षेपेण ।

पूर्णे स्याताम् अन्तराशिकाभ्यां तदा हरांशो भवतीश्रमस्य ॥ १ ॥

The mathematical procedure is as follows: given a non-square number n , to find $\sqrt{n}$:

1. Choose an initial approximation a_1 (the “approximate value”)
2. Compute the residue: $r = n - a_1^2$
3. Form the next approximation: $a_2 = a_1 + \frac{r}{2a_1}$
4. Iterate for successively better approximations

कल्प्यते—

(१) ०, अ, अ', अ'',

(२) १, सो, सो', सो'',

(३) अ, क_१, क_२, क_३, क_४,

अत्र क_१, क_२, क_३, —अयववदासन्नमूलस्यासन्नमानानि $\frac{प}{ल}$, $\frac{प'}{ल'}$, $\frac{प''}{ल''}$ चेति ।

$$\sqrt{n} = \frac{\frac{\sqrt{n+१}}{१} \cdot १ + १}{\frac{\sqrt{n+१}}{१} \cdot १ + १} = \frac{१(\sqrt{n+१}) + १ \cdot १}{१(\sqrt{n+१}) + १ \cdot १}$$

4.4 Final Concluding Verses (pp. 166–168)

The treatise concludes with verses that reflect on the nature of mathematical knowledge and pay homage to the tradition:

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ

बीजगणिताध्यायः समाप्तः ॥

अस्ति वैराशिकं पाठी बीजं च विमला मतिः ।
किमन्यत् सुखदीनामतो मन्दाऽर्थमुच्यते ॥

गणककर्मान्तरमयं बाललीलावनोगम्यं
सकलगणितसारं सोपपट्टिप्रकारम् ।
इति बहुगुणयुक्तं सर्वदोषैर्विमुक्तं
पठ पठ मतिबुद्ध्यै लाघवं प्रौढिसिद्धयै ॥

“There is the study of this work and pure intellect — what else [is needed]? Thus is declared for the delight of the wise. This [work], accessible as a child’s play [yet] the essence of all mathematics with demonstrations, endowed with many virtues, free from all faults — read, read with intelligence and understanding, for ease and the attainment of maturity.”

The author reflects on the completion of the work:

तथा गोले मयोक्तम् ।
उत्थ सदसतां मर्तानां वैराशिकमात्रमेव पाठो बुद्धिरेव बीजम् ।
तथा गोलाख्याय मयोक्तम् ॥

“Thus I have spoken of the sphere. For people ascending from the unreal to the real, the mere reading [of this text] is the seed of intelligence. Thus I have spoken of what is called the sphere.”

Colophon

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

Editor’s Note (p. 168)

वि०—इति कूपाङ्कनमुखाकरे यद्विह भास्करबीजमपूर्वकम् ।
तदुपनिषमतीव चमत्कृतिं विभिधरस्य चकार च कारणम् ॥

“That unprecedented algebra of Bhāskara, which here [appears] at the mouth of the well-digit-mine, having ascertained it with the Upaniṣad [of commentary], became the cause of the appearance of manifold wonder.”

Publisher’s Catalogue (Chowkhamba Sanskrit Book Depot)

The final two pages contain a catalogue of Sanskrit publications available from the CHOWKHAMBA SANSKRIT BOOK DEPOT (Chowkhamba Griham), 4015 Thatheri Bazar, Varanasi. The catalogue lists works across all branches of Sanskrit learning, including:

No.	Title	Price	Pages
19	Golaprakāśasya Golādhyāya-Bījagaṇita-vāsiṣṭha-koṇa-gaṇite Śrī-Candraśekhāra-saṃskṛtite	...	☒
20	Golaprakāśīya-cāpīyatrikōṇa-gaṇitasya “Vivāha” ṭīkā Śrīmahāvira-pañḍita-viracitā	...	☒
21	Vāsanāvyañjana-ṭīkā Śrīgaṇeśa-dhara-mitra-kṛtā		

No.	Title	Price	Pages
☐	ṭikā-sahitā	...	☐
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